MIL-H-4081B
21 August 1969
SUPERSEDING
MIL-H-4081A
2 March 1953

MILITARY SPECIFICATION

HOLDERS: COUNTERSINK, ADJUSTABLE, MICROMETER STOP AND CUTTERS, COUNTERSINK

This specification is mandatory for use by all departments and agencies of the Department of Defense

*1. SCOPE AND CLASSIFICATION

*1.1 Scope. This specification covers holders and cutters for countersinking ferrous and non-ferrous materials.

*1.2 Classification. Holders and cutters shall be of the following type, class and size as specified (see 6.2).

* Holders:

Type I	Internal taper spindle
Type II	Internal threaded spindle

* Cutters:

Class 1	Countersink, plain taper shank
Class 2	Countersink, external threaded shank

Size	Body Diameter	Pilot Diameter
A	0.437	0.093
B	0.437	0.125
C	0.437	0.156
D	0.437	0.187
E	0.500	0.125
F	0.500	0.250
G	0.625	0.312

*2. APPLICABLE DOCUMENTS

*2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

* Federal

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

QQ-T-570	Tool Steel, Alloy
QQ-W-470	Wire, Steel, Carbon, Spring, Music

Military

MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals
MIL-H-15424	Hand Tools, Packaging of

STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability

Military

MIL-STD-105	Sampling Procedure and Tables of Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook

H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

American Society for Testing and Materials (ASTM)

ASTM E18	Rockwell Hardness and Rockwell Superficial Hardness for Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

United States of America Standard (USAS)

USAS B46.1	Surface Roughness, Waviness and Lay

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N. Y. 10016.)

3. REQUIREMENTS

*3.1 Material. Chemical composition and hardenability of steel components for the holder assembly shall be equal to the requirements of FED-STD-66 for AISI B1113 steel, except that material for the knockout pin, the spindle and the helical spring(s) shall be as specified in the application paragraphs herein.

*3.2 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to exclude holders and cutters which otherwise comply with the requirements of this specification.

*3.3 Threads. All threads shall be in the inch system and shall conform to class 2 tolerance limits as defined in Handbook H28.

*3.4 Treatment and finish.

*3.4.1 Heat treatment. Heat treat hardness requirements specified herein shall be Rockwell "C" scale values. Hardness vertification shall be in accordance with ASTM E18 test procedure.

*3.4.2 Surface texture. All surface roughness limits specified herein are arithmetical average (AA) microinch values of waviness and lay as defined in USAS B46.1.

*3.4.3 Protective coating. All ferrous metal components of the holder assembly, except the bearing contact areas and the cutter mounting and centering surfaces on the spindle, shall be black oxide coated in accordance with MIL-C-13924.

*3.5 Construction. Holders shall be type I or type 2 as specified (see 1.2) and similar to figure 1 or figure 2. The assembly shall consist of a spring loaded self-retracting spindle, ball or roller type thrust bearing, oil impregnated radial bearing, adjusting screw, adjustable skirt, skirt adjusting sleeve and sleeve locknut. The holder length shall be adjustable and shall have locking devices and retainers to pervent unintentional adjustment, disengagement or separation of parts. The spindle travel length shall not be less than 9/32 inch. The oil impregnated radial bearing may be an integral part of the threaded adjusting screw or may be a separate sleeve type replaceable bearing.

*3.5.1 Spindles. The spindles shall be of one piece construction, heat treated to 41-44 hardness, with provisions for mounting the cutter on one end and a chucking shank on the opposite end. The cutter mounting taper, for type I holder, and cutter centering angle, for type II holder, shall be concentric with the bearing contact areas to 0.003 inch total indicator reading (TIR). The chucking shank shall be 0.250 $\pm$ 0.002 inch diameter by 3/4 inch long with a fillet radius blend into the body diameter. Surface texture on the bearing contact areas and cutter centering surface shall not be rougher than 32 microinch (AA).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

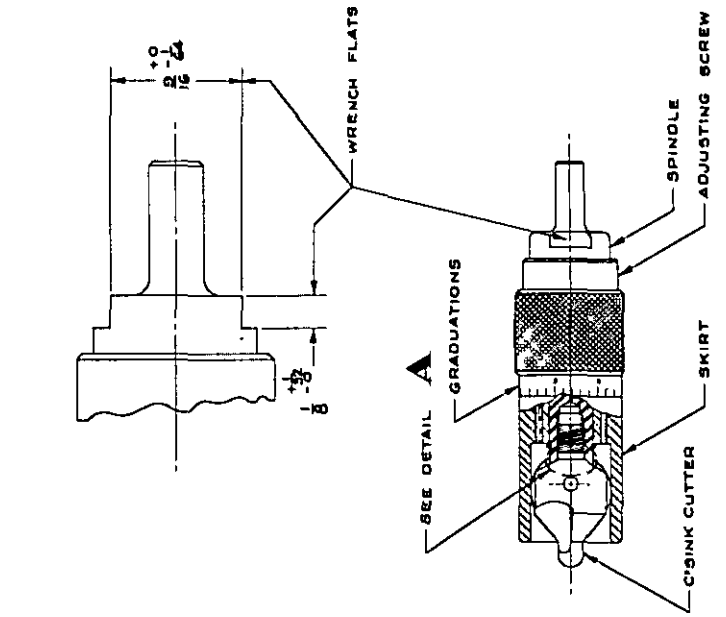


FIGURE 2
TYPE II

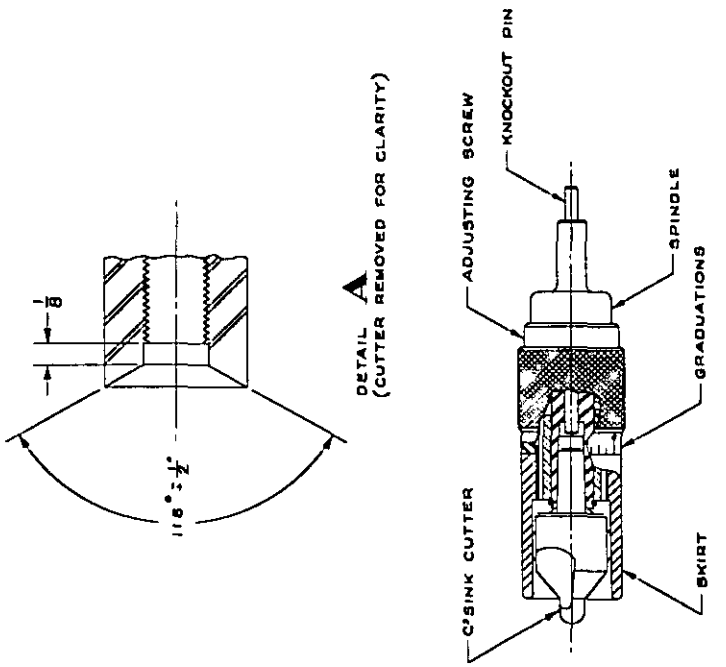


FIGURE 1
TYPE I

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

*3.5.1.1 Spindle for type I holder. The internal self-locking cutter mounting taper shall be 0.250 ± 0.001 inch diameter at large end with 0.600 + 0.00-0.005 inch per foot taper. Knockout pin hole through the spindle shall be 0.129 ± 0.005 inch diameter (see figure 1).

*3.5.1.2 Spindle for type II holder. Cutter mounting end of the spindle shall be threaded 1/4 - 28 UNF-2B by 1/2 inch deep. Cutter centering angle at the thread entrance shall be 118 degrees ± 1/2 degree included angle by 0.340 inch diameter. Wrenching flats shall be provided for installing and removing cutters (see figure 2).

*3.5.2 Knockout pin. The knockout pin shall be made of carbon steel, not less than 0.125 inch diameter and shall be heat treated to 45.50 Rockwell C hardness.

*3.5.3 Spring, spindle retracting. Spring wire shall be equal to the requirements of QQ-W-470. Diameter, length and compression resistance shall be the manufacturer's standard.

*3.5.4 Thrust bearing. The bearing shall be installed to absorb spindle thrust load in both the cutting and retracted positions.

*3.5.5 Adjusting screw. The screw shall provide for overall holder length adjustment. The external thread shall mate with the internal threaded portion of the skirt, adjusting sleeve and the sleeve locknut. The screw shall house the spindle, thrust bearing, retracting spring and, when applicable, the sleeve type radial bearing.

*3.5.6 Skirt. The internal thread at one end of the skirt shall provide for cutter depth control adjustment. Chip clearance opening, beginning not more than 3/16 inch from the footplate face, shall be not less than 75 percent of the skirt circumference. A permanent index line, marked with a zero (0) indicator, shall be clearly visible over the entire adjustment range.

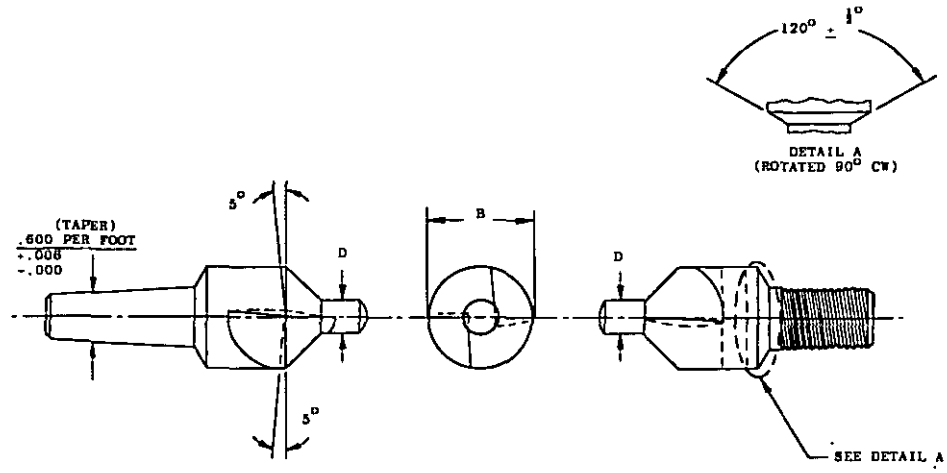
*3.5.7 Adjusting sleeve and locknut. The outside diameter of the sleeve and locknut shall be knurled, ribbed or beaded to enable adjustment without the use of special tools or wrenching devices. Outside diameter, at both ends, of the sleeves shall not be larger than the root diameter of the knurled, ribbed or beaded portion for a length distance not less than 3/16 inch and not rougher than 125 microinch (AA) Circumference at the skirt end shall be permanently marked to indicate cutter depth adjustment in increments of 0.001 inch and shall be marked with progressive numerical characters at every fifth increment; (i.e., 5-10-15 etc.).

*3.6 Cutters. The cutters shall be made of high speed tool steel and shall be heat treated to 60-63 hardness. The pilot diameter, taper shank or cutter centering angle, as applicable, and the cutting edges shall be concentric to 0.003 inch TIR. Class, size and countersink angle (see 1.2 and Table I) shall be as specified (see 6.2).

*3.6.1 Class 1 cutter. Configuration and dimensional requirements shall be as shown in figure 3 and as specified in Table I.

*3.6.2 Class 2 cutter. Configuration and dimensional requirements shall be as shown in figure 4 and as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.



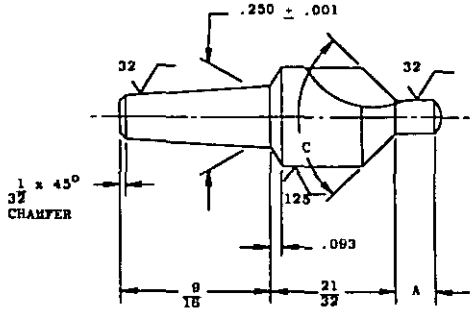


FIGURE 3
CLASS 1

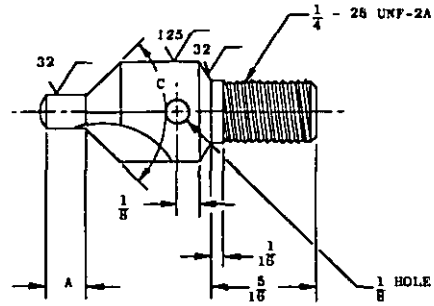


FIGURE 4
CLASS 2

TABLE I

SIZE	A +- 1/64	B +- .005	C ± 0° 30'						D +.000 -.001
			60°	78°	82°	100°	110°	115°	
A	.3/32	.437							.093
B	1/8	.437							.125
C	5/32	.437							.156
D	3/16	.437							.187
E	1/18	.500							.125
F	1/4	.500							.250
G	5/16	.625							.312

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

*3.7 Identification of product. Each holder and cutter shall be marked for identification in accordance with MIL-STD-130. Holders shall be marked with the specification number, type and the manufacturers name or part number. Cutter marking shall include the class, size, countersink included angle and the manufacturers name or trademark.

*4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Certified report. The supplier shall furnish a certified test report showing that the holders and cutters satisfactorily conform to this specification. The report shall include a statement that the holders and cutters did pass the dimensional inspection, all tests in section 4 and shall indicate the actual cutter hardness range and the applicable production lot number.

*4.2 Classification of examination and tests. The examinations and tests required to assure conformance of those items or lots of items offered for acceptance are classified as quality conformance.

*4.3 Lot definition.

*4.3.1 Production lot. A production lot shall consist of all holders or cutters manufactured by the same process, heat treated in the same manner and produced as one continuous run or order.

*4.3.2 Inspection lot. All holders or cutters of the same type, class and size taken from a production lot and offered for delivery at one time shall be considered a lot for purposes of inspection and tests.

*4.3.3 Sampling for quality conformance inspection. Samples shall be taken from the inspection lot in accordance with MIL-STD-105, based on single sampling plan for normal inspection, at the level specified. Defects in excess of the acceptance number shall be cause for rejection of the representative lot.

*4.4 Test method.

*4.4.1 Visual and dimensional inspection. Each sample holder and cutter shall be visually and dimensionally inspected for conformance with section 3. Classification of defects shall be as specified in table II. Sampling shall be at inspection level I with an acceptable quality level (AQL) of 2.5 percent defective for major defects and 6.5 percent defective for total defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

TABLE II
CLASSIFICATION OF DEFECTS

Category	Defects
Major:	
101	Holder dimensions, configuration and concentricity not as specified.
102	Inoperative or binding spindle in lateral or radial movement or both.
103	Depth control adjustment (incremental or entire range) require use of wrenching devices.
104	Incremental divisions and numerical marking on sleeve, illegible, inaccurate or not permanent.
105	Cutter hardness, countersink angle, pilot diameter and concentricity not as specified.
Minor:	
201	Finish; not as specified; evidence of fins, burns or sharp edges.
202	Marking; not as specified; illegible, or not permanent.

*4.4.2 Functional test. Each sample holder shall be tested for freedom of spindle rotation, locknut and retainer effectiveness and stability. Mount the chucking shank in a rotating spindle and bring the skirt footplate in firm contact with a smooth metal surface. Advance the holder spindle approximately 1/4 inch, for firm spindle pre-load and positive footplate pressure, to simulate actual operating condition. Rotate the spindle at 750 and 1,000 RPM respectively for not less than 15 minutes at each spindle speed. Excessive heating, spindle seizure, rotation of footplate, unintentional loosing of locknut or disengagement of the spindle retainer shall be cause for rejection of the representative lot. Sampling shall be at inspection level I with an AQL of 2.5 percent defective.

*4.4.3 Spindle concentricity and adjustment accuracy. After successful completion of tests in 4.4.2 mount each holder in a stationary fixture and inspect the spindle concentricity and cutter depth control accuracy. The internal taper or cutter centering angle, as applicable, shall be concentric with the chucking shank to 0.003 inch TIR. Adjust the skirt (cutter depth control) at 0.005 inch increments as marked on the adjusting sleeve for one complete revolution. Actual deviation at each 0.005 inch adjustment shall not exceed 0.001 inch (non-cumulative) with not more than 0.005 inch total deviation for one complete revolution. Total cutter depth control adjustment shall be at least 9/32 inch. Sampling shall be at inspection level II with an AQL of 2.5 percent defective.

*4.4.4 Performance test. Each sample holder and cutter of the type style and size specified in the contract or order shall be subjected to the performance test. The countersink cutting test shall be performed with the holder mounted

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

in a portable, electrical or pneumatic powered, hand tool. Holders and cutters shall be subjected to not less than ten repetitive countersink operations in stainless steel sheet, type 302, 8 gage minimum thickness. The countersink shall be at least 0.406 ± 0.005 inch major diameter for all holes and the countersink surfaces shall not be rougher than 32(AA). The included countersink angle and the relation of angle to the reference plane shall be accurate to ± 1/2 degree. Sampling shall be at inspection level II with an AQL of 1.5 percent defective.

*4.5 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to insure compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. The holders and cutters shall be cleaned, preserved, packaged, packed and marked for shipment in accordance with the provisions of MIL-H-15424. Levels of preservation, packaging, and packing shall be as specified in the contract or order (see 6.2).

6. NOTES

*6.1 Intended use. Holders and cutters manufactured in accordance with this specification are intended for use with power driven portable hand tools or stationary machines to prepare mounting holes for countersink head fasteners.

*6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

* b. Type holder required (see 1.2).

c. Cutter class, size and included countersink angle required (see 1.2, 3.6 and Table I).

d. Selection of applicable levels of preservation, packaging, packing and marking (see 5.1).

*6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contents irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

*Custodians

Air Force - 84

DSA - IP

Reviewer

Air Force - 84

Army - GL

User Activity

Army - EL, AV

Air Force - 84

Preparing Activity

Air Force - 84

Project Number

5133-0030

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AFZ-WP-O-MAY 64 505

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:47:47
Check the source to verify that this is the current version before use.